

LAB EXPERIMENT RECORD

Set 10: Survey Responses Analysis

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze survey response data using a grouped bar chart, stacked bar chart, and table in R, and to build an interactive R Shiny dashboard combining all three.

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

PROGRAM 1: Grouped Bar Chart for Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

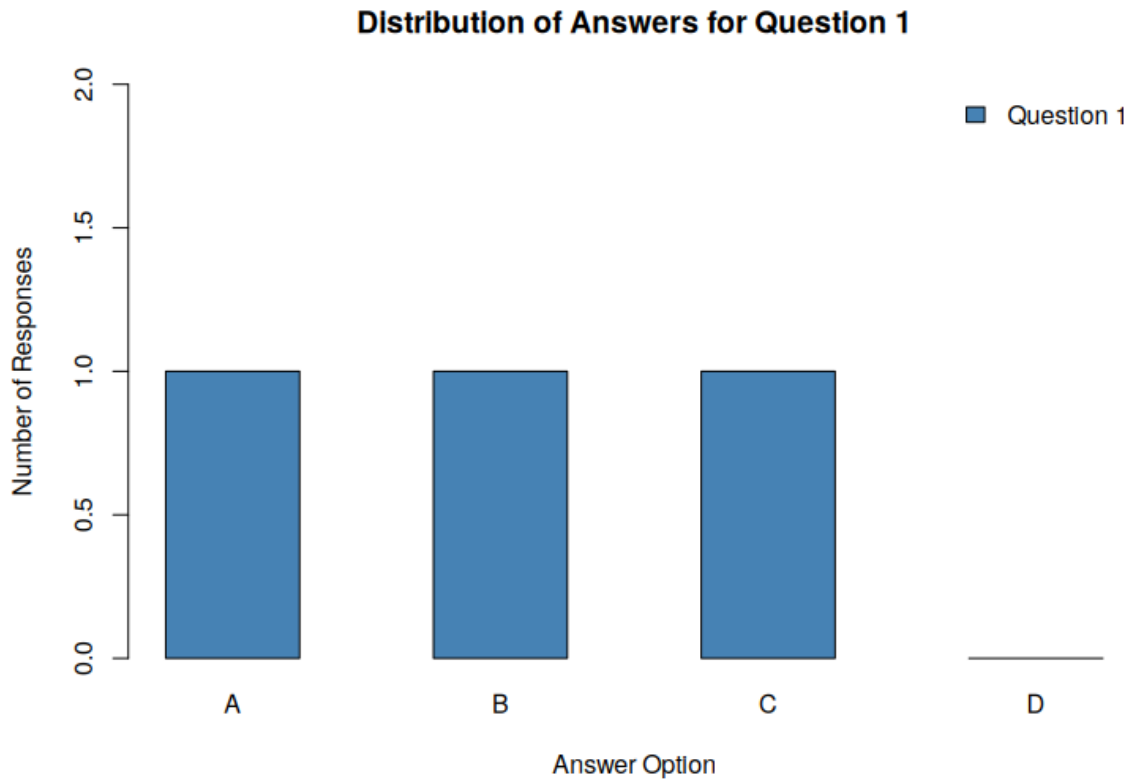
```
# Create dataset for Set 10
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey_data)

# Count answers for Question 1 (all answer options A to D)
q1_counts <- table(factor(survey_data$Question_1, levels = c("A", "B", "C", "D")))
q1_matrix <- matrix(q1_counts, nrow = 1,
  dimnames = list("Question 1", names(q1_counts)))
print(q1_matrix)

# 1. Grouped Bar Chart for Question 1
barplot(q1_matrix,
  beside = TRUE,
  col = "steelblue",
  xlab = "Answer Option",
  ylab = "Number of Responses",
```

```
main = "Distribution of Answers for Question 1",  
ylim = c(0, 2),  
legend.text = rownames(q1_matrix),  
args.legend = list(x = "topright", bty = "n"))
```



Output 1: Distribution of Answers for Question 1

Observation: For Question 1, options A, B, and C were each chosen by one respondent, and no respondent chose option D. The answers are evenly spread, so there is no single preferred answer for Question 1.

PROGRAM 2: Stacked Bar Chart of All Responses

Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
# Create dataset for Set 10
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

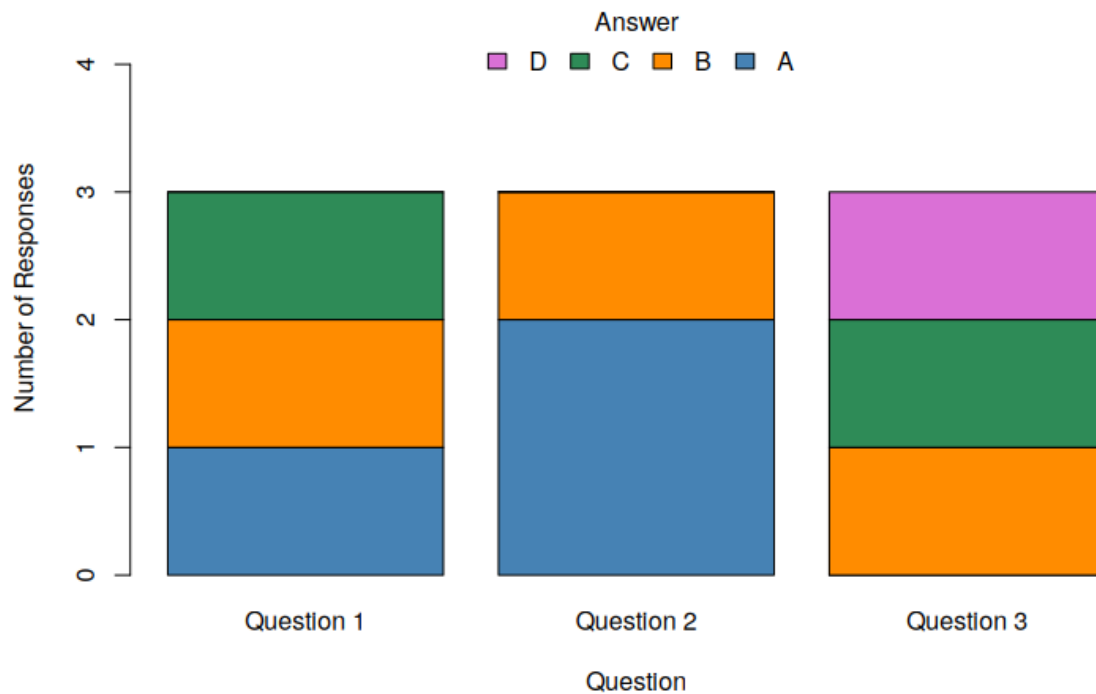
print(survey_data)

# Combine all answers into long format
responses <- data.frame(
  Question = rep(c("Question 1", "Question 2", "Question 3"), each = 3),
  Answer = c(survey_data$Question_1, survey_data$Question_2, survey_data$Question_3)
)

# Answer counts for each question
response_table <- table(factor(responses$Answer, levels = c("A", "B", "C", "D")),
  responses$Question)
print(response_table)

# 2. Stacked Bar Chart of Responses for All Questions
barplot(response_table,
  col = c("steelblue", "darkorange", "seagreen", "orchid"),
  xlab = "Question",
  ylab = "Number of Responses",
  main = "Overall Distribution of Survey Responses",
  ylim = c(0, 4.5),
  legend.text = TRUE,
  args.legend = list(title = "Answer", x = "top", horiz = TRUE, bty = "n"))
```

Overall Distribution of Survey Responses



Output 2: Overall Distribution of Survey Responses

Observation: Each question received three responses. Question 2 is dominated by option A (2 of 3 responses), while Questions 1 and 3 are spread across three different options. Option D was chosen only once, in Question 3. Across all questions, A and B are the most common answers (3 times each).

PROGRAM 3: Table of Survey Responses

Build a table to show the survey response data for each survey. Label the table elements.

```
# Create dataset for Set 10
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)

print(survey_data)

# 3. Table of survey responses
survey_table <- survey_data
colnames(survey_table) <- c("Survey ID", "Question 1", "Question 2", "Question 3")

cat("=== SURVEY RESPONSES TABLE ===\n")
print(survey_table, row.names = FALSE)
```

```
=== SURVEY RESPONSES TABLE ===
> print(survey_table, row.names = FALSE)
  Survey ID Question 1 Question 2 Question 3
        1          A          B          C
        2          B          A          D
        3          C          A          B
>
```

Output 3: Survey Responses Table

Observation: The table shows the answer given in each survey for all three questions with clearly labelled columns, so each respondent's full set of answers can be read in one row.

PROGRAM 4: Interactive Dashboard for Survey Responses

Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart, and the table for interactive exploration of survey responses.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 10
survey_data <- data.frame(
  Survey_ID = c(1, 2, 3),
  Question_1 = c("A", "B", "C"),
  Question_2 = c("B", "A", "A"),
  Question_3 = c("C", "D", "B")
)
options <- c("A", "B", "C", "D")
ans_cols <- c("steelblue", "darkorange", "seagreen", "orchid")

ui <- fluidPage(
  titlePanel("Survey Responses Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("surveys", "Survey ID:", choices = 1:3, selected = 1:3)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("groupPlot", height = "300px")),
        column(6, plotOutput("stackPlot", height = "300px"))
      ),
      h4("Survey Responses Table"),
      tableOutput("surveyTable")
    )
  )
)

server <- function(input, output) {
  filtered <- reactive({ subset(survey_data, Survey_ID %in% input$surveys) })

  output$groupPlot <- renderPlot({
    q1 <- table(factor(filtered()$Question_1, levels = options))
    barplot(matrix(q1, nrow = 1, dimnames = list("Question 1", options)), beside = TRUE,
      col = "steelblue", ylim = c(0, 2), xlab = "Answer Option",
      ylab = "Number of Responses", main = "Answers for Question 1")
  }, width = 420, height = 300, res = 96)

  output$stackPlot <- renderPlot({
    d <- filtered()
    tab <- sapply(c("Question_1", "Question_2", "Question_3"),
      function(q) table(factor(d[[q]], levels = options)))
    colnames(tab) <- c("Question 1", "Question 2", "Question 3")
    barplot(tab, col = ans_cols, ylim = c(0, 4.5), xlab = "Question",
      ylab = "Number of Responses", main = "Overall Distribution of Responses",
      legend.text = options, args.legend = list(x = "top", horiz = TRUE, bty = "n"))
  }, width = 420, height = 300, res = 96)
```

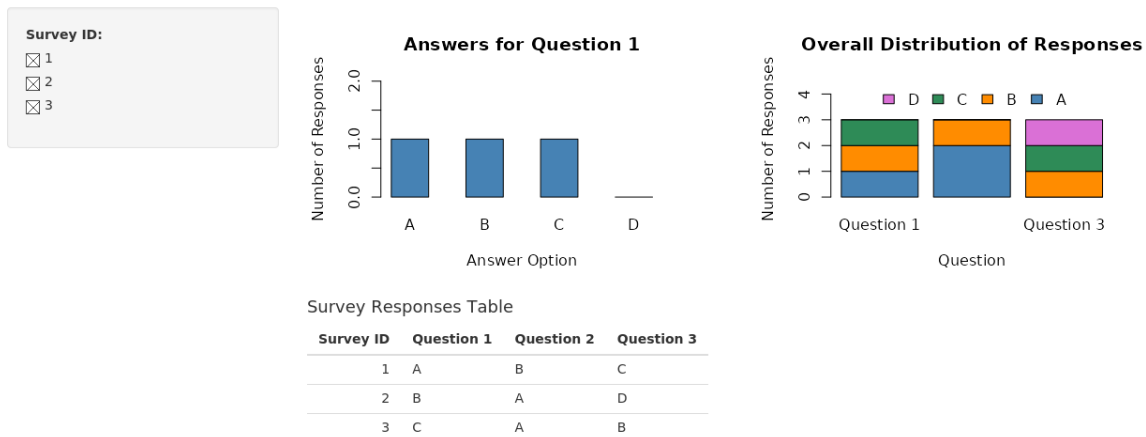
```

output$surveyTable <- renderTable({
  d <- filtered()
  colnames(d) <- c("Survey ID", "Question 1", "Question 2", "Question 3")
  d
}, digits = 0)
}

shinyApp(ui = ui, server = server)

```

Survey Responses Dashboard



Output 4: Survey Responses Interactive Dashboard (R Shiny)

Observation: The dashboard shows the Question 1 grouped bar chart, the overall stacked bar chart, and the survey table together. Unticking a Survey ID removes that survey's answers from both charts and the table at once.

Procedure

1. Create the survey data frame with Survey ID and the answers to Questions 1, 2, and 3.
2. Count the Question 1 answers with `table()` and `factor()` and draw the grouped bar chart using `barplot()` with `beside = TRUE`.
3. Combine all answers into long format, build an answer $\times$ question table, and use `barplot()` to draw the stacked bar chart.
4. Rename the columns with `colnames()` and print the labelled response table.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts and table.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The survey response data was successfully visualized using a grouped bar chart, stacked bar chart, and table in R, and an interactive R Shiny dashboard was built combining all three views.

Key R Functions Used

Function	Purpose
<code>data.frame()</code>	Creates the survey dataset.
<code>factor()</code> / <code>table()</code>	Counts answers including options with zero responses.
<code>matrix()</code>	Arranges the Question 1 counts for the grouped bar chart.
<code>barplot()</code>	Creates the grouped and stacked bar charts.
<code>rep()</code>	Builds the long-format question column.
<code>colnames()</code> / <code>print()</code>	Labels and displays the response table.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>renderTable()</code>	Displays the filtered table on the dashboard.
<code>shinyApp()</code>	Runs the interactive dashboard.